IntelliTwin: Asymmetric, Condition-Adaptive Conformal Prediction for Industrial Turbofan Predictive Maintenance

Research Dissertation — Comprehensive Working Draft

Pre-Registered Methodology and Experimental Protocol (Frozen Commit `bb9e34579343881a831e3f052608ee15e183df76`)

Final Empirical Results Blinded Pending Experimental Execution (Gates UPD-02, UPD-03, UPD-04)

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Institutional Declaration and Submission Notice

Administrative Status Notice: *In accordance with research documentation governance, all statutory declarations, certificates of originality, supervisor approval certificates, and institutional degree award specifications remain explicitly designated* `PENDING_INSTITUTIONAL_SPECIFICATION`*. No signatures, supervisor approvals, or institutional endorsements have been fabricated.*

This volume represents an evidence-linked, pre-registered academic dissertation draft establishing the theoretical foundation, methodological architecture, calibration pipelines, and confirmatory statistical protocols.

Abstract

Predictive maintenance for safety-critical aviation assets requires balancing the catastrophic safety and financial hazards of unplanned in-flight engine failures against the economic waste of premature component retirement. Conventional prognostic models focus on minimizing cycle-level Remaining Useful Life (RUL) regression errors without quantifying predictive uncertainty or accommodating asymmetric maintenance loss functions.

In this dissertation, we develop and formalize **IntelliTwin**, an end-to-end framework integrating data-driven degradation predictors (Gradient Boosted Regressors and recurrent Long Short-Term Memory networks) with hierarchical, decision-aware conformal prediction. The framework enforces strict engine-level disjoint data partitioning (304 training, 122 calibration, 61 validation, and 122 held-out test engines across NASA C-MAPSS sub-datasets FD001, FD002, and FD004), causal sliding temporal windows ($w = 30$, stride 1), exact rational engine-weighted empirical quantiles, and condition-stratified conformal calibration with deterministic fallback rules. Nonconformity scoring incorporates an asymmetric loss parameter ($\beta = -\ln 2$) that penalizes dangerous RUL overestimation.

Downstream operational utility is evaluated through discrete-event fleet maintenance simulation. To determine whether observed cost benefits stem from calibrated uncertainty rather than simple over-conservatism, IntelliTwin is evaluated against four named controls across three baseline families (matched-width heuristic intervals, widened intervals, point safety margins, and earlier fixed interventions). Confirmatory hypothesis testing is governed by paired engine-level cluster bootstrap inference ($B = 10{,}000$) with pre-registered practical-effect margins and Bonferroni family-wise error rate control (98.75% confidence intervals).

Provisional Status Notice: The empirical entries reported on held-out test splits are currently blinded awaiting completion of software verification and experimental execution gates (UPD-02, UPD-03, UPD-04).

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Chapter 1

Introduction

1.1 Industrial Context and Problem Statement

Aviation and industrial turbofan engines are complex aeromechanical systems that operate under extreme thermo-mechanical stresses, severe vibrational loads, and dynamically varying flight envelopes [1]. In contemporary fleet operations, ensuring operational reliability while managing lifecycle maintenance expenditures represents a paramount economic and safety challenge. Unplanned in-flight engine shutdowns, emergency diversions, and catastrophic component structural failures impose severe safety risks and immense financial costs, frequently exceeding scheduled maintenance overhauls by orders of magnitude [2].

Historically, commercial aviation maintenance relied predominantly on conservative time-based preventative policies, replacing subassemblies at predetermined flight-hour or cycle milestones regardless of actual physical health [3]. While safe, this schedule-driven paradigm leads to premature replacement of functional hardware, wasting significant operational life.

With the emergence of Internet of Things (IoT) sensor telemetry and high-capacity flight data recorders, Condition-Based Maintenance (CBM) and Predictive Maintenance (PdM) paradigms have become foundational to next-generation aviation operations. These paradigms seek to predict the Remaining Useful Life (RUL) of critical components dynamically from multivariate sensor streams, enabling opportunistic scheduling before physical failure occurs [4].

1.2 The Prognostic-to-Decision Gap

Despite significant advances in deep learning for degradation modeling, an acute methodological gap persists between cycle-level machine learning accuracy and downstream operational utility:

1. **Cycle-Level Error Optimization**: Point regression architectures—such as Long Short-Term Memory (LSTM) networks, Temporal Convolutional Networks (TCNs), and Gradient Boosted Regressors (GBRs)—are conventionally trained to minimize symmetric cycle-level loss functions such as Mean Squared Error (MSE) or Mean Absolute Error (MAE) [1,5]. These metrics treat underestimating and overestimating RUL symmetrically.

2. **Asymmetric Cost Realities**: In real-world aviation logistics, overestimating RUL by 10 cycles can allow an engine to operate beyond its physical failure boundary, causing catastrophic damage. In contrast, underestimating RUL by 10 cycles merely results in premature maintenance and a minor loss of usable component life [1, 6].

3. **Uncalibrated Uncertainty**: When probabilistic or interval outputs are produced via heuristic dispersion or Bayesian approximations, they typically rely on Gaussianity or asymptotic normality assumptions that fail under non-stationary degradation dynamics and regime shifts [7]. Uncalibrated prediction intervals provide no statistical guarantees on coverage.

1.3 The IntelliTwin Approach

To resolve this gap, the **IntelliTwin** research prototype integrates data-driven base degradation predictors with hierarchical, decision-aware conformal prediction theory. Hierarchical conformal theory provides marginal predictive coverage under its stated sampling and exchangeability assumptions. IntelliTwin uses an engine-weighted construction motivated by that theory. Applicability of those assumptions to ordered, lifetime-stratified degradation trajectories is not established merely by weighting engines equally; reliability is therefore evaluated empirically under the frozen protocol.

Furthermore, the framework introduces two critical adaptations:

- **Regime-Stratified Calibration**: Conformal calibration is conditioned on operational flight regime clusters, preventing severe conditional undercoverage in rare or stressed operating conditions.

- **Decision-Aware Asymmetry**: Nonconformity scores are scaled via an asymmetric parameter ($\beta = -\ln 2$), tilting prediction intervals to guard against hazardous RUL overestimation.

1.4 Research Objectives and Scope

This dissertation formalizes four pre-registered research questions:

- **RQ1 (Sequential Utility)**: Can decision-aware conformal calibration improve sequential maintenance cost efficiency while preserving nominal coverage guarantees?
- **RQ2 (Subgroup Reliability)**: Does regime-stratified calibration eliminate worst-case coverage deficits across operational clusters without excessive interval inflation?
- **RQ3 (Attribution vs. Matched Controls)**: Are observed cost benefits attributable to calibrated uncertainty rather than simple, ad-hoc conservatism?
- **RQ4 (Robustness & Generalization)**: Does the calibration advantage generalize across diverse base predictors (GBR, LSTM), operating regimes, and degradation modes?

1.5 Thesis Organization

The remainder of this dissertation is organized as follows:

- Chapter 2 provides a comprehensive review of turbofan prognostics, conformal prediction, and maintenance decision theory.
- Chapter 3 details the mathematical problem formulation and authoritative hypothesis framework.
- Chapter 4 presents the complete methodological pipeline from causal windowing to asymmetric calibration.
- Chapter 5 outlines the software architecture, engine-level partitioning, and provenance tracking.
- Chapter 6 formalizes the confirmatory experimental setup, baselines, and bootstrap inference.
- Chapter 7 presents the status of experimental verification and discussion of expected outcomes.
- Chapter 8 discusses threats to validity, research boundaries, and conclusions.
- Appendix A provides the complete reproducibility protocol and execution commands.

Chapter 2

Literature Review and Related Work

2.1 Introduction to the Literature Landscape

Prognostics and Health Management (PHM) for mission-critical aeromechanical assets has evolved dramatically over the past two decades. The progression from classical physics-based fatigue models to modern data-driven deep learning architectures reflects increasing sensor telemetry availability and algorithmic sophistication. However, the operational adoption of data-driven Remaining Useful Life (RUL) predictors in aviation maintenance scheduling remains severely constrained by two fundamental deficiencies: the absence of rigorous, distribution-free uncertainty quantification, and the disconnect between cycle-level regression accuracy and downstream maintenance decision utility.

This chapter presents a structured synthesis of the literature across five thematic pillars:

1. Turbofan Remaining Useful Life (RUL) prediction paradigms;
2. Uncertainty quantification and conformal calibration;
3. Operating regime heterogeneity and subgroup reliability;
4. Maintenance decision theory and conservatism controls;
5. The specific research gap addressed by the IntelliTwin framework.

2.2 Pillar 1: Turbofan Remaining Useful Life Prediction

The foundational benchmark for data-driven turbofan degradation research is the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset introduced by Saxena et al. [1]. C-MAPSS models thermodynamic degradation in com-

mercial turbofans under varying flight profiles, providing run-to-failure trajectories for high-pressure compressor and fan deterioration.

Early algorithmic approaches focused on classical machine learning, such as Support Vector Regression (SVR), Random Forests, and Gradient Boosted Decision Trees (GBDT) [4]. With the deep learning paradigm shift, recurrent neural architectures became the prevailing standard. Recurrent architectures, such as Long Short-Term Memory (LSTM) networks and attention mechanisms [5], have been widely applied to capture temporal degradation dynamics in multivariate telemetry alongside convolutional networks [2].

However, across the turbofan prognostics literature, model evaluation has remained almost universally confined to symmetric error metrics, specifically Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), or ad-hoc asymmetric exponential scoring functions [1]. These point-prediction paradigms fail to provide decision-makers with confidence bounds necessary for safety-critical maintenance dispatch.

2.3 Pillar 2: Uncertainty Quantification and Conformal Prediction

To address the limitations of point estimates, various uncertainty quantification (UQ) methodologies have been explored in PHM:

- **Bayesian Neural Networks (BNNs)**: Employ Monte Carlo dropout or variational inference to estimate weight posterior distributions. While theoretically principled, BNNs impose heavy computational overheads and rely heavily on prior distribution specifications.

- **Quantile Regression**: Estimates conditional quantiles directly via the pinball loss. However, quantile regression provides no finite-sample coverage guarantees and is vulnerable to quantile crossing under non-stationary degradation.

- **Conformal Prediction**: Conformal prediction theory provides a distribution-free uncertainty framework [7]. By evaluating nonconformity scores on a reserved calibration dataset, inductive split-conformal inference constructs prediction intervals targeting valid finite-sample marginal coverage:

$$P(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha. \tag{2.1}$$

Recently, Javanmardi et al. [7] and Yang et al. [8] explored conformal prediction for industrial prognostics. However, standard split-conformal methods assume exchangeabil-

ity, which is challenged by non-stationary degradation trajectories and multi-regime flight envelopes.

2.4 Pillar 3: Operating Regime Heterogeneity and Subgroup Reliability

In real-world aviation operations, engines traverse diverse flight regimes (climb, cruise, descent, sea-level take-off). In C-MAPSS sub-datasets FD002 and FD004, six discrete operating regimes induce severe sensor non-stationarity.

When conformal calibration is applied globally across a pooled calibration split, the resulting marginal guarantee $1 - \alpha$ guarantees only aggregate coverage across the entire dataset. Lee et al. [9] and Benabdennour et al. [10] demonstrated that global calibration frequently yields severe conditional undercoverage in rare or high-stress operational clusters. While hierarchical conformal prediction (HCP) allows conditioning on group indicators, naive partitioning across small subgroup samples can induce catastrophic quantile variance. Establishing structured fallback protocols for sparse clusters is essential for industrial viability.

2.5 Pillar 4: Maintenance Decisions and Conservatism Controls

The ultimate objective of prognostic modeling is to inform maintenance scheduling. In operational fleet logistics, an asymmetric cost structure prevails:

$$C_{\text{unplanned}} \gg C_{\text{preventive}}. \tag{2.2}$$

Overestimating RUL allows an engine to operate until catastrophic functional failure, incurring heavy safety penalties, whereas underestimating RUL merely causes premature scheduled replacement, forfeiting a fraction of usable life [1, 6].

Cortes-Gomez et al. [11] introduced utility-directed conformal calibration, while Robinson et al. [6] explored risk-aware RUL bounds. However, an unresolved question in the literature is whether observed maintenance cost reductions stem from genuine uncertainty calibration or merely from trivial over-conservatism (i.e., triggering maintenance earlier regardless of prediction accuracy). Resolving this attribution requires evaluating calibrated intervals against rigorously matched-width heuristic baselines, widened intervals, and fixed safety margins.

2.6 Pillar 5: The IntelliTwin Synthesis and Research Gap

Despite the individual advancements cataloged above, no prior study has simultaneously established:

1. Complete engine-level data leakage isolation without cycle-level shuffling;
2. Condition-stratified conformal calibration with deterministic fallback for rare flight regimes;
3. Asymmetric nonconformity scaling aligned with maintenance cost structures;
4. End-to-end discrete-event maintenance simulation evaluating sequential fleet utility;
5. Attribution testing against matched-conservatism controls under pre-registered cluster-bootstrap statistical protocols.

IntelliTwin directly resolves this research gap through an open, reproducible, and pre-registered experimental framework.

Chapter 3

Problem Formulation and Research Hypotheses

3.1 Mathematical Problem Formulation

Consider a fleet of N turbofan engines indexed by $i \in \{1, \ldots, N\}$. Each engine i is monitored over discrete operational cycles $t \in \{1, \ldots, T_i\}$, where $T_i \in \mathbb{N}$ denotes the terminal run-to-failure cycle of the unit. At each cycle t, an engine emits an operational state vector:

$$\mathbf{s}_{i,t} = \left[u_{i,t,1}, u_{i,t,2}, u_{i,t,3}, s_{i,t,1}, \ldots, s_{i,t,21}\right]^\top \in \mathbb{R}^{24}, \tag{3.1}$$

consisting of 3 ambient flight condition settings and 21 thermodynamic sensor measurements. Following training-only qualification and variance screening, seven uninformative sensor channels are eliminated, resulting in a standardized 17-dimensional state vector $\mathbf{x}_{i,t} \in \mathbb{R}^{17}$.

3.1.1 Causal History and Information Horizon

To enforce physical temporal causality and prevent future-information leakage, predictions at cycle t must condition exclusively on a fixed backward temporal window of length $w = 30$ cycles:

$$\mathbf{H}_{i,t} = [\mathbf{x}_{i,t-w+1}, \ldots, \mathbf{x}_{i,t}] \in \mathbb{R}^{w \times 17}, \quad t \geq w. \tag{3.2}$$

Trajectories are sampled with unit stride ($s = 1$) and strict avoidance of zero-padding. Consequently, no prognostic predictions are issued for cycles $t < w$.

3.1.2 Target Degradation Variable

The physical Remaining Useful Life (RUL) of engine i at cycle t is defined as:

$$y_{i,t} = T_i - t, \quad t \in [w, T_i]. \tag{3.3}$$

During base model training, an upper piecewise cap $R_{\max} = 125$ cycles is applied to reflect negligible initial degradation:

$$y_{i,t}^{\text{train}} = \min(y_{i,t}, R_{\max}). \tag{3.4}$$

In calibration, validation, and held-out test evaluations, the target remains strictly uncapped ($y_{i,t}$) to evaluate true physical end-of-life accuracy.

3.2 Sequential Maintenance Decision Problem

At each cycle $t \geq w$, the prognostic digital twin generates a point estimate $p_{i,t} \geq 0$ and a calibrated lower prediction bound $L_{i,t}$. The maintenance policy π evaluates whether to maintain immediately:

$$a_{i,t} = \begin{cases} \text{MAINTAIN_NOW} & \text{if } L_{i,t} \leq 1.0 \text{ cycle}, \\ \text{CONTINUE} & \text{otherwise}, \end{cases} \tag{3.5}$$

Actions occur at cycles $t \in [30, T_i - 1]$. Preventive maintenance occurs immediately upon trigger and terminates the episode; there is no intervention lead time.

If replacement is triggered at cycle $t < T_i$, scheduled preventive maintenance occurs immediately, incurring cost C_{PM} and leaving unused (wasted) operational life $W_i = T_i - t$. If the engine reaches its true terminal cycle T_i without scheduled maintenance, in-service failure occurs, incurring penalty C_{FAIL} with wasted life $W_i = 0$.

The total simulated maintenance cost for engine i is given by:

$$\text{Cost}_i = \begin{cases} C_{\text{PM}} + C_{\text{WASTE}} \cdot (T_i - t) & \text{if maintenance occurs at } t < T_i, \\ C_{\text{FAIL}} & \text{if failure occurs at } t = T_i. \end{cases} \tag{3.6}$$

The study specifies three synthetic economic scenarios:

- **`primary_reference`**: $C_{\text{PM}} = 1.0\,\text{SCU}, C_{\text{FAIL}} = 5.0\,\text{SCU}, C_{\text{WASTE}} = 0.20\,\text{SCU/cycle}$.
- **`failure_sensitive`**: $C_{\text{PM}} = 1.0\,\text{SCU}, C_{\text{FAIL}} = 10.0\,\text{SCU}, C_{\text{WASTE}} = 0.20\,\text{SCU/cycle}$.
- **`waste_sensitive`**: $C_{\text{PM}} = 1.0$ SCU, $C_{\text{FAIL}} = 5.0$ SCU, $C_{\text{WASTE}} = 0.40$ SCU/cycle.

Offline interval evaluation includes all qualified endpoints 30 through T_i, independently of policy stopping. The equality values (20, 45, and 10 cycles) compare preventive-plus-waste cost with failure cost, and do not represent policy decision thresholds.

3.3 Authoritative Research Hypotheses

The research questions, null and alternative hypotheses, primary outcome metrics, practical-effect thresholds, and statistical decision rules are permanently locked under Architecture Decision Record ADR-018. The authoritative specification is reproduced in its entirety below.

Governance Rules on Hypothesis Integrity (ADR-018)

> **Authoritative Precedence:** This document is a derived LaTeX rendering. The YAML specification (`hypothesis_framework.yaml`) and statistical protocol (`statistical_inference_protocol.md`) remain the primary immutable authorities.
>
> **Decision Outcome Rules:** Each research question possesses a pre-specified primary outcome, comparator set, frozen numerical practical-effect threshold, and cluster-bootstrap decision rule classifying results into a three-way outcome: **Supported**, **Unsupported**, or **Inconclusive**.
>
> **Outcome Hierarchy:** The primary outcome determines the confirmatory claim for each research question. Secondary outcomes verify operational constraints, reliability trade-offs, and failure safeguards. A favorable secondary outcome cannot replace failure on the primary outcome.
>
> **No Post-Hoc Alteration:** Hypotheses, comparators, subgroups, thresholds, and decision rules are frozen and must not be altered after observing confirmatory test data. Any post-hoc modification must be explicitly designated as an exploratory analysis.
>
> **Statistical Sample Unit:** The independent statistical unit is the physical **engine unit** ($i \in \{1, \ldots, N\}$). Individual operational cycles (t) within an engine trajectory are autocorrelated and must **NEVER** be treated as independent statistical observations.

Numerical Evaluation Contract (Protocol `step06-v2`)

- **Primary Maintenance Cost Scenario**: `primary_reference` ($C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$ SCU per engine).

- **Secondary Sensitivity Scenarios**: `failure_sensitive` ($C_{\text{FAIL}} = 10.0$) and `waste_sensitive` ($C_{\text{WASTE}} = 0.40$).
- **Nominal Coverage**: Primary confirmatory coverage 90% ($\alpha = 0.10$); descriptive sensitivities at 80% and 95%.
- **Cost Improvement Margin**: $\Delta_{\text{cost}} > 0.20$ SCU per engine.
- **Coverage Error Improvement Margin**: $\Delta_{\text{error}} > 0.01$ (one percentage point error reduction).
- **Maximum Absolute Coverage Error Ceiling**: $E < 0.05$ (prespecified 85%–95% empirical acceptance band).
- **Interval Noninferiority Margins**: $D_{\text{width}} < 1.0$ cycle for MPIW and $D_{\text{score}} < 1.0$ cycle for Winkler interval score.
- **Robustness Non-Harm Margins**: $\Delta_{\text{cost}} > -0.20$ SCU per engine and $\Delta_{\text{error}} > -0.01$ in every cell.
- **Failure Safeguard**: Observed fleet failure rate increase $\Delta_{\text{fail}} \leq 0.0$ against comparator.
- **Fixed Model Seeds**: Five prespecified training seeds: `[2026, 2027, 2028, 2029, 2030]`.

Statistical Resampling and Multiplicity Control

1. **Resampling Engine**: 10,000 paired engine-cluster bootstrap resamples with replacement, stratified by dataset lifetime quartiles (Q_1–Q_4) fixed from split manifests.
2. **PRNG Specification**: `numpy.random.Generator` backed by `PCG64` with seed sequence `SeedSequence([2026, 6, dataset_number])`.
3. **Decision Confidence Level**: Confirmatory decision rules use **98.75% cluster-bootstrap percentile intervals** (quantiles 0.00625 and 0.99375), providing Bonferroni family-wise error rate control ($\alpha = 0.05/4 = 0.0125$) across the four primary research questions. Descriptive 95% intervals (quantiles 0.025 and 0.975) are reported for context.
4. **Subgroup Eligibility & Bound Propagation**: Subgroup eligibility (≥ 10 engines) is frozen before resampling. If an eligible subgroup is absent in replicate b, its absolute error is bounded by $[0.0, 0.90]$ at 90% nominal coverage, propagating conservatively.

5. **Intersection-Union Testing (IUT)**: An RQ is Supported if and only if **every required component condition passes** at the 98.75% decision interval level.

Formal Hypothesis Definitions

Research Question 1 (RQ1): Sequential Maintenance Utility and Coverage Reliability

- **Question**: *Can a decision-aware, condition-adaptive conformal RUL calibration method improve sequential maintenance utility while preserving reliable prediction-interval coverage?*
- **Null Hypothesis ($H_{0,1}$)**: The proposed method does not improve sequential maintenance utility by at least 0.20 SCU per engine over both global and standard condition-aware calibration, or fails to maintain required coverage reliability.
- **Alternative Hypothesis ($H_{1,1}$)**: The proposed method improves maintenance utility by at least 0.20 SCU per engine over both global and standard condition-aware calibration while maintaining required coverage reliability.
- **Primary Outcome**: Total simulated maintenance cost per engine contrast ($\Delta_{\text{cost}} = \overline{\text{Cost}}_B - \overline{\text{Cost}}_P$).
- **Primary Comparators**: Global conformal calibration (`global_hcp`) and standard condition-aware calibration (`regime_hcp`) on C-MAPSS FD002 held-out test split under `primary_reference` at 90% nominal coverage (both GBR and LSTM required).
- **Supported Logic**: For both GBR and LSTM, lower 98.75% CI of $\Delta_{\text{cost}} > 0.20$ SCU against both baselines, upper 98.75% CI of marginal, worst-regime, and worst-stage coverage error < 0.05, and observed fleet failure rate increase ≤ 0.0 against both baselines.
- **Unsupported Logic**: For either model, upper 98.75% CI of $\Delta_{\text{cost}} < 0.20$ SCU against either baseline, lower 98.75% CI of any coverage error > 0.05, or observed fleet failure rate increases over either baseline.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Research Question 2 (RQ2): Conditional Subgroup Reliability and Interval Sharpness

- **Question**: *Can the proposed method improve reliability across operating regimes and degradation conditions without excessively widening prediction intervals?*
- **Null Hypothesis ($H_{0,2}$)**: The proposed method does not improve worst-group conditional coverage error by at least 0.01 over both baselines, or causes interval width or interval score to deteriorate by 1.0 cycle or more.
- **Alternative Hypothesis ($H_{1,2}$)**: The proposed method improves worst-group conditional coverage error by at least 0.01 over both baselines while restricting interval width and score deterioration to under 1.0 cycle.
- **Primary Outcome**: Worst-group conditional coverage error contrast ($\Delta_{\text{error}} = \text{WorstErr}_B - \text{WorstErr}_P$).
- **Required Subgroup Families**: Operating regimes (C_1–C_6) and degradation life-cycle stages (Early, Mid, Late thirds).
- **Supported Logic**: For both models and both subgroup families, lower 98.75% CI of $\Delta_{\text{error}} > 0.01$ against both baselines, upper 98.75% CI of worst-group error < 0.05, upper 98.75% CI of $D_{\text{width}} < 1.0$ cycle, and upper 98.75% CI of $D_{\text{score}} < 1.0$ cycle.
- **Unsupported Logic**: For either model or subgroup family, upper 98.75% CI of $\Delta_{\text{error}} < 0.01$ against either baseline, lower 98.75% CI of worst-group error > 0.05, lower 98.75% CI of $D_{\text{width}} > 1.0$ cycle, or lower 98.75% CI of $D_{\text{score}} > 1.0$ cycle.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Research Question 3 (RQ3): Decision-Value Attribution vs. Matched Controls

- **Question**: *Are the maintenance benefits of the proposed method attributable to decision-aware uncertainty calibration rather than simply more conservative prediction intervals or earlier maintenance?*
- **Null Hypothesis ($H_{0,3}$)**: The proposed method does not improve maintenance utility by at least 0.20 SCU per engine over every prespecified matched-conservatism control family, or violates coverage constraints.

- **Alternative Hypothesis ($H_{1,3}$)**: The proposed method provides maintenance improvements beyond those achievable through matched-conservatism baselines, demonstrating useful decision information rather than simple over-conservatism.
- **Primary Outcome**: Maintenance cost improvement over matched-conservatism controls ($\Delta_{\text{cost}} = \overline{\text{Cost}}_{\text{control}} - \overline{\text{Cost}}_{P}$).
- **Four Evaluated Control Configurations across Three Families**:
  1. `matched_width_interval`: λ-scaled baseline interval matching proposed validation MPIW within ≤ 1.0 cycle (GBR: `global_hcp`, $\lambda = 0.92$; LSTM: `regime_hcp`, $\lambda = 0.90$).
  2. `widened_interval`: Parameterized interval expansion $\lambda \geq 1.0$ (GBR: `regime_hcp`, $\lambda = 1.00$; LSTM: `global_hcp`, $\lambda = 1.00$).
  3. `point_safety_margin`: Thresholding $p \leq 1 + m$ (GBR: $m = 5$; LSTM: $m = 4$).
  4. `earlier_intervention`: Elevated thresholding $L \leq h$ (GBR: `regime_hcp`, $h = 1$; LSTM: `global_hcp`, $h = 1$).
- **Supported Logic**: For both GBR and LSTM, lower 98.75% CI of $\Delta_{\text{cost}} > 0.20$ SCU against **every single prespecified control configuration**, upper 98.75% CI of marginal, regime, and stage coverage error < 0.05, and observed fleet failure rate increase ≤ 0.0 against all controls.
- **Unsupported Logic**: For either model, upper 98.75% CI of $\Delta_{\text{cost}} < 0.20$ SCU against any control configuration, lower 98.75% CI of any coverage error > 0.05, or fleet failure rate increases over any control.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Research Question 4 (RQ4): Robustness and Generalization

- **Question**: *Does the proposed method retain its calibration and maintenance-decision benefits across operating conditions, base RUL predictors, datasets, and repeated experimental runs?*
- **Null Hypothesis ($H_{0,4}$)**: The proposed method fails to achieve aggregate improvements ($\Delta_{\text{cost}} > 0.20$ SCU, $\Delta_{\text{error}} > 0.01$) across the evaluation matrix, or exhibits material harm ($\Delta_{\text{cost}} < -0.20$ SCU or $\Delta_{\text{error}} < -0.01$) in any individual configuration cell.

- **Alternative Hypothesis ($H_{1,4}$)**: The proposed method achieves aggregate improvements across the evaluation matrix without material harm in any individual configuration cell.
- **Evaluation Matrix**: Six mandatory cells: FD001 × GBR, FD001 × LSTM, FD002 × GBR, FD002 × LSTM, FD004 × GBR, FD004 × LSTM, weighted equally at $w_k = 1/6$ per cell.
- **Supported Logic**: Lower 98.75% CI of weighted aggregate $\Delta_{\text{cost}} > 0.20$ SCU, lower 98.75% CI of weighted aggregate $\Delta_{\text{error}} > 0.01$ for regimes and stages, lower 98.75% CI of $\Delta_{\text{cost}} > -0.20$ SCU in every cell, lower 98.75% CI of $\Delta_{\text{error}} > -0.01$ in every cell, and all cell safeguards hold across all 6 cells.
- **Unsupported Logic**: Upper 98.75% CI of aggregate $\Delta_{\text{cost}} < 0.20$ SCU or aggregate $\Delta_{\text{error}} < 0.01$, or upper 98.75% CI of cell $\Delta_{\text{cost}} < -0.20$ SCU or $\Delta_{\text{error}} < -0.01$ in any individual cell, or any cell safeguard fails.
- **Inconclusive Logic**: No condition fails, but at least one decision interval covers its boundary, or inference is degenerate.

Summary Evaluation Contract Matrix

ID	Experiment	Primary Outcome	Key Comparators	Statistical Unit	Practical Threshold (98.75% CI)
RQ1	Experiment A	Maintenance cost per engine (Δ_{cost})	`global_hcp` & `regime_hcp`	Engine ($N = 52$)	$\Delta_{\text{cost}} > 0.20$ SCU per engine
RQ2	Experiment B	Worst-group coverage error (Δ_{error})	`global_hcp` & `regime_hcp`	Engine ($N = 52$)	$\Delta_{\text{error}} > 0.01$; $D_{\text{width}}, D_{\text{score}} < 1.0$
RQ3	Experiment C	Utility vs. matched controls (Δ_{cost})	4 named matched controls	Engine ($N = 52$)	$\Delta_{\text{cost}} > 0.20$ SCU vs. all controls
RQ4	Experiment D	Weighted aggregate & cell non-harm	6 cells ($w_k = 1/6$ each)	Engine ($N = 122$)	Agg: $> 0.20, > 0.01$; Harm: $> -0.20, > -0.01$

Table 3.1: IntelliTwin Confirmatory Hypotheses, Metrics, Comparators, and Frozen Thresholds (ADR-018).

Chapter 4

Methodology

4.1 Methodological Architecture Overview

The IntelliTwin methodological framework links machine learning degradation modeling, distribution-free uncertainty calibration, and sequential decision simulation into a cohesive prognostic pipeline. The pipeline operates under three immutable architectural principles:

1. **Causal Temporal Isolation**: Predictions condition strictly on past operational history within a 30-cycle causal horizon; zero-padding and future lookahead are strictly precluded.

2. **Engine-Level Disjoint Calibration**: Conformal nonconformity scores are computed exclusively on dedicated, reserved calibration engine trajectories that share no engine identities with training or held-out test splits.

3. **Asymmetric Cost Alignment**: Nonconformity metrics modulate intervals to reflect the severe cost penalties of component failure relative to preventive replacement.

4.2 Data Preprocessing and Causal Windowing

4.2.1 Training-Only Z-Score Normalization

Preprocessing uses training-only z-score normalization. Means and population standard deviations are estimated with `ddof=0`. An exactly zero standard deviation is replaced by an effective scale of 1.0, while the mean is still subtracted:

$$x_{\text{normalized}} = \frac{x - \mu_{\text{train}}}{s_{\text{train}}} \tag{4.1}$$

where $s_{\text{train}} = \sigma_{\text{train}}$ when $\sigma_{\text{train}} > 0$, and $s_{\text{train}} = 1.0$ otherwise.

Dataset-specific processing executes strictly prior to window construction:

- **FD001**: Single nominal operating condition; global training-fitted z-score normalization across all qualified features.
- **FD002 and FD004**: Operational flight settings are standardized using training data; six operating regimes (C_1–C_6) are learned from standardized training settings using K-Means clustering ($k = 6$, random_state = 2026, n_init = 10). Canonical regime labels are assigned based on lexicographic centroid ordering. Sensor normalization parameters are fitted separately within each training regime.

4.2.2 Feature Space Construction

The pipeline enforces a frozen 17-feature selection supported by the recorded qualification procedure:

`setting1, setting2, setting3, s2, s3, s4, s7, s8, s9, s11, s12, s13, s14, s15, s17, s20, s21`

Seven sensors ($s_1, s_5, s_6, s_{10}, s_{16}, s_{18}, s_{19}$) are excluded under this recorded qualification procedure; we do not claim they are mathematically constant or universally uninformative across all datasets and regimes.

4.2.3 Causal Windowing Protocol

Temporal feature matrices are formed via rolling causal windows of duration $w = 30$ operational cycles with unit stride ($s = 1$), strictly avoiding zero-padding:

$$\mathbf{W}_{i,t} = [x_{i,t-29}, x_{i,t-28}, \ldots, x_{i,t}] \in \mathbb{R}^{30\times 17}, \quad t \geq 30. \tag{4.2}$$

Windows are formed exclusively within continuous engine trajectories; windows never span across engine boundaries, and cycles $t < 30$ are unpredicted. In training splits, target RUL is capped at $\min(\text{rul_true}, 125)$ cycles as a modeling convention. Point predictions are evaluated as $p = \max(0, \text{raw_prediction})$ without upper clipping at 125.

4.3 Base Degradation Predictors

We examine two distinct algorithmic paradigms receiving identical 30-cycle causal information horizons:

1. **Gradient Boosted Regressor (GBR)**: `sklearn.ensemble.HistGradientBoostingRegressor` operating on C-order flattened 30×17 feature vectors (510 features). Winning configurations selected on validation data across seeds 2026–2027: `gbr_lr010_leaf15_l2_0` for FD001, and `gbr_lr010_leaf31_l2_0` for FD002 and FD004.

2. **Long Short-Term Memory (LSTM)**: Single-layer unidirectional PyTorch LSTM (`input_size=17`, `hidden_size=64`, `num_layers=1`, `batch_first=True`, `bidirectional=False`, internal LSTM `dropout=0.0`), extracting the final sequence output, followed by `nn.Dropout(p=0.20)` and `nn.Linear(64, 1)`. Winning configuration: `lstm_h64_lr0010` (learning rate $\eta = 0.0010$), using early stopping on validation uncapped-RUL RMSE.

Each selected configuration is trained across five final research seeds (`2026, 2027, 2028, 2029, 2030`), yielding 30 frozen checkpoints in `experiments/base_models/model_registry.json`. We do not call the five GBR seeds independent experiments, nor assert universal deterministic behavior beyond the recorded runs.

4.4 Conformal Calibration Formulation

4.4.1 Engine-Weighted Quantile Computation

Standard pooled conformal prediction introduces survivor bias in industrial run-to-failure telemetry because long-lived engines contribute more sliding windows than short-lived units. IntelliTwin implements hierarchical engine-weighted quantiles motivated by Lee, Barber, and Willett [9].

For FINAL calibration with m contributing engines, where engine i contains $n_i = T_i - 29$ valid endpoints, the empirical distribution function is:

$$F_{\text{final}}(q) = \frac{1}{m+1} \sum_{i=1}^{m} \frac{1}{n_i} \sum_{t=1}^{n_i} \mathbb{I}(\text{score}_{it} \leq q) \tag{4.3}$$

The mass of each finite endpoint is $\frac{1}{(m+1)n_i}$. The total finite mass is $\frac{m}{m+1}$, with the remaining probability mass $\frac{1}{m+1}$ placed at $+\infty$. The conformal threshold q^* for nominal coverage $1-\alpha$ is the smallest observed score whose CDF reaches $1-\alpha$. If no finite score reaches it, $q^* = +\infty$. (In contrast, normalized development empirical CDFs use $1/m$ weighting when selecting configurations via leave-one-engine-out cross-validation).

4.4.2 Conformal Calibration Variants

Let $p \geq 0$ denote non-negative predicted RUL in cycles, $y = y \geq 0$ denote uncapped ground-truth RUL, and $g \in \{C_1, \ldots, C_6\}$ denote the observed operational regime.

1. **Global HCP (`global_hcp`)**: Evaluates absolute residual nonconformity score$(p, y) = |y - p|$. The resulting prediction interval is:

$$L = \max(0,\ p - q_{\text{global}}), \quad U = p + q_{\text{global}} \tag{4.4}$$

2. **Regime-Stratified HCP (`regime_hcp`)**: Evaluates absolute residuals, fitting separate regime-specific thresholds q_g. Regime support requires $m_g \geq \max(10, \lceil 1/\alpha \rceil - 1)$ distinct contributing calibration engines in that regime (10 engines for 80% and 90% coverage; 19 engines for 95% coverage). When m_g is insufficient, the method falls back deterministically to q_{global} fitted for the same method and coverage level, recording fallback.

3. **Decision-Aware HCP (`decision_aware_hcp`)**: Incorporates asymmetric penalty scaling:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \tag{4.5}$$

$$a(p) = \sigma(p)\exp(\beta z(p)), \quad b(p) = \sigma(p)\exp(-\beta z(p)) \tag{4.6}$$

where $\gamma \in \{0, 1, 2\}$ and $\beta \in \{-\ln 2, 0, +\ln 2\}$. Nonconformity scores and inverted intervals are:

$$\text{score}(p, y) = \max\left(\frac{p - y}{a(p)},\ \frac{y - p}{b(p)},\ 0\right) \tag{4.7}$$

$$L = \max(0,\ p - q_g \cdot a(p)), \quad U = p + q_g \cdot b(p) \tag{4.8}$$

4.4.3 Mathematical Mechanics of Negative Beta ($\beta < 0$)

At fixed p, q, and γ, negative β decreases $a(p)$ relative to $\beta = 0$ and increases $b(p)$. The lower bound L is nondecreasing and the upper bound U is nondecreasing. We do not claim $a < 1$ or $b > 1$ for every p, as the $\sigma(p)$ factor also influences their magnitude. Furthermore, this mathematical fact does not establish improved maintenance utility or earlier intervention; nonconformity thresholds q^* are fitted separately and utility must be evaluated empirically in sequential simulation.

Candidate `gamma2_betam1` ($\gamma = 2, \beta = -\ln 2$) was selected as the optimal feasible configuration across all six mandatory cells and frozen under ADR-021.

4.5 Discrete-Event Maintenance Simulation

The sequential maintenance policy executes step-by-step across unseen test engine trajectories. At cycle $t \in [30, T_i - 1]$, if the lower prediction bound satisfies $L(t) \leq 1.0$ cycle,

the policy issues a `MAINTAIN_NOW` command. Preventive maintenance occurs immediately upon decision and terminates the episode (zero intervention lead time).

If maintenance occurs at $t < T_i$, cost is:

$$\text{Cost} = C_{\text{PM}} + C_{\text{WASTE}} \cdot (T_i - t) \tag{4.9}$$

leaving unused operational life $W_i = T_i - t$. If the engine reaches its true physical failure cycle T_i without maintenance, in-service failure occurs, incurring cost $\text{Cost} = C_{\text{FAIL}}$ with wasted life $W_i = 0$.

The study evaluates three synthetic cost scenarios:

- **`primary_reference`**: $C_{\text{PM}} = 1.0\,\text{SCU}, C_{\text{FAIL}} = 5.0\,\text{SCU}, C_{\text{WASTE}} = 0.20\,\text{SCU/cycle}$.
- **`failure_sensitive`**: $C_{\text{PM}} = 1.0\,\text{SCU}, C_{\text{FAIL}} = 10.0\,\text{SCU}, C_{\text{WASTE}} = 0.20\,\text{SCU/cycle}$.
- **`waste_sensitive`**: $C_{\text{PM}} = 1.0$ SCU, $C_{\text{FAIL}} = 5.0$ SCU, $C_{\text{WASTE}} = 0.40$ SCU/cycle.

Offline interval evaluation includes all qualified endpoints 30 through T_i, independently of policy stopping. The equality values (20, 45, and 10 cycles) compare preventive-plus-waste cost with failure cost, and do not represent policy decision thresholds.

Chapter 5

Software Architecture and Implementation

5.1 System Architecture and Package Structure

The IntelliTwin framework is implemented as a modular Python package (`intellitwin`) structured to guarantee strict segregation of concerns, end-to-end provenance tracking, and scientific reproducibility. The core package is organized into dedicated sub-packages:

- `intellitwin.data`: Implements C-MAPSS dataset loaders, feature qualification routines, training-only z-score normalization, and causal sliding window generators ($w = 30$, stride 1, no padding). Enforces immutable engine-level partitioning based on pre-generated split manifests.
- `intellitwin.models`: Implements `sklearn.ensemble.HistGradientBoostingRegressor` for Gradient Boosted Regressors (GBR) and PyTorch modules for recurrent Long Short-Term Memory (LSTM) networks. Standardizes model serialization, inference pipelines, and CPU execution verification.
- `intellitwin.uncertainty`: Implements inductive split-conformal calibrators, exact rational engine-weighted quantile computation, operating-regime clustering, and asymmetric nonconformity scaling ($\beta = -\ln 2$). Enforces deterministic fallback rules for regimes with insufficient engine counts ($m_g < \max(10, \lceil 1/\alpha \rceil - 1)$).
- `intellitwin.simulator`: Implements the discrete-event sequential maintenance simulator, modeling continuous degradation progression, immediate preventive overhauls upon trigger, and in-service failure penalties.
- `intellitwin.decision`: Contains maintenance decision policy classes, executing rule-based triggers on point predictions, heuristic safety margins, and conformal lower bounds ($L(t) \leq 1.0$ cycle).

- `intellitwin.evaluation`: Contains cluster-bootstrap statistical inference engines (10,000 paired engine resamples), metric calculators (RMSE, MAE, coverage, MPIW, Winkler score, maintenance cost, failure rates), and the automated Intersection-Union Test (IUT) evaluation logic.
- `intellitwin.provenance`: Manages cryptographically verifiable SHA-256 artifact hashing, git commit tracking, environment profiling, and execution metadata capture.

5.2 Registry Architecture and Frozen Artifacts

To maintain strict governance over research artifacts, all model checkpoints, calibrator bundles, and experimental parameters are tracked via immutable JSON registries:

1. **Base Model Registry** (`experiments/base_models/model_registry.json`): Catalogs the 30 base model checkpoints across 6 experimental cells (FD001, FD002, FD004 × GBR, LSTM) and 5 official seeds (2026–2030), including file paths, model architecture specifications, training metrics, and SHA-256 hashes.
2. **Calibration Registry** (`experiments/calibration/calibration_registry.json`): Documents 90 fitted conformal calibrator bundles covering global, regime-stratified, and decision-aware variants.
3. **Selection Freeze** (`experiments/calibration/selection/selection_freeze.json`): Records the hyperparameter configuration selected strictly on validation engine splits (`gamma2_betam1`) and frozen prior to held-out test evaluation.
4. **Protocol Freeze** (`experiments/pretest_verification_v2/protocol_freeze.json`): Formally locks the evaluation parameters, cost constants, seed sequences, bootstrap sample counts (10,000), and decision thresholds for confirmatory testing.
5. **Control Reconstruction** (`experiments/pretest_verification_v2/control_reconstructio`: Catalogs the exact selection parameters for the four pre-registered matched-conservatism controls selected on FD002 validation engines across seeds 2026–2030.

5.3 Provenance Tracking and Auditability

Every official experimental execution generates an audit record capturing stored fields precisely:

- Timestamp in UTC ISO format (`timestamp_utc`);

- Python runtime version string (`python_version`) and host OS platform string (`platform`);
- Compute device indicator (`device`, defaulted strictly to `cpu`);
- Core installed library versions: `numpy`, `pandas`, `scikit_learn`, `torch`, and `pyyaml`;
- Exact command arguments executed (`command_args`);
- Git commit SHA and working-tree dirty status;
- Input data file SHA-256 digests and output artifact checksums.

We do not claim that individual runs record CPU hardware model, physical memory capacity, a full lockfile, or full internal PRNG state.

5.4 Committed Pre-Test Audit Findings (Commit `bb9e345`)

At reviewed commit `bb9e34579343881a831e3f052608ee15e183df76`, pretest audit artifacts establish:

- `step08_full_audit.json` reports all 6 mandatory audit checks executed and passed;
- 60 base-model partition evaluations (covering the `validation` and `calibration` partitions across all 30 models) reproduce with maximum absolute prediction difference of 0.0;
- `pretest_audit.json` confirms status `PASS` and `PRETEST_READY`;
- Baseline manifest distinguishes 652 tracked, protected Git artifacts from 72 historical local-only search checkpoints;
- Official held-out evaluation status remains `NOT_STARTED`, and all confirmatory research question outcomes remain `PENDING`.

5.5 Demarcation of Implemented Modules vs. Planned UI

To maintain rigorous scientific boundaries, we explicitly demarcate implemented research components from planned presentation features:

- **Implemented Research Prototype**: All data ingestion, feature qualification, base model training, conformal calibration, discrete-event maintenance simulation, and bootstrap inference modules are fully implemented, tested, and audited in `src/intellitwin/`.

- **Planned Demonstrator UI**: An interactive web-based dashboard for replaying engine degradation trajectories and visualizing simulated fleet maintenance schedules is planned for final presentation integration (tracked under `UPD-06`). The demonstrator will operate strictly in offline replay mode and makes no claim of real-time physical engine telemetry integration or flight certification.

Chapter 6

Experimental Design and Confirmatory Protocol

6.1 Benchmark Datasets and Stratification

The empirical evaluation of IntelliTwin is conducted across three sub-datasets of the NASA C-MAPSS turbofan engine degradation benchmark [1]:

1. **FD001**: Single operating condition (sea level), single failure mode (high-pressure compressor degradation), comprising 100 complete run-to-failure engine units.
2. **FD002**: Six operating flight conditions, single failure mode (HPC degradation), comprising 260 engine units.
3. **FD004**: Six operating flight conditions, two concurrent failure modes (HPC and fan degradation), comprising 249 engine units.

6.2 Engine-Level Disjoint Partitioning

To prevent data leakage across temporal cycles, engine trajectories are partitioned into disjoint splits using master seed 2026 prior to any data preprocessing:

- **FD001** (100 engines): 50 training (50%), 20 calibration (20%), 10 validation (10%), and 20 held-out test engines (20%).
- **FD002** (260 engines): 130 training (50%), 52 calibration (20%), 26 validation (10%), and 52 held-out test engines (20%), stratified across engine lifetime quartiles (Q_1–Q_4).
- **FD004** (249 engines): 124 training (50%), 50 calibration (20%), 25 validation (10%), and 50 held-out test engines (20%).

Across the complete benchmark suite, 304 engines are allocated to training, 122 to calibration, 61 to validation, and 122 to final held-out confirmatory evaluation (total 609 engines).

6.3 Experimental Cells and Replications

The experimental matrix comprises a 3×2 grid (6 experimental cells):

$$\{\text{FD001}, \text{FD002}, \text{FD004}\} \times \{\text{GBR}, \text{LSTM}\}. \tag{6.1}$$

Two point-prediction architectures operate on identical 30-cycle causal horizons:

1. **Gradient Boosted Regressor (GBR)**: `sklearn.ensemble.HistGradientBoostingRegressor` operating on C-order flattened $30 \times 17 = 510$-dimensional vectors. Selected configs: `gbr_lr010_leaf15_l2_0` (FD001) and `gbr_lr010_leaf31_l2_0` (FD002, FD004).
2. **Long Short-Term Memory (LSTM)**: Single-layer unidirectional PyTorch LSTM ($H = 64$, recurrent dropout 0.0, followed by `nn.Dropout(p=0.20)` and `nn.Linear(64, 1)`, learning rate $\eta = 0.0010$, `lstm_h64_lr0010`) with early stopping on validation uncapped RMSE.

Each cell is trained across five pre-registered random seeds:

$$\mathcal{S} = \{2026, 2027, 2028, 2029, 2030\}. \tag{6.2}$$

This produces 30 base model checkpoints and 90 conformal calibrator bundles. Two selection seeds (2026 and 2027) used during model development are distinguished from the five final model seeds. We do not characterize the five GBR seeds as independent experiments, nor assert universal deterministic behavior beyond the recorded runs.

6.4 Matched-Conservatism Control Baselines

To isolate the operational decision value of calibrated uncertainty from trivial policy over-conservatism (RQ3), we formulate four named controls across three distinct baseline families:

1. **Matched-Width Interval Family (`matched_width_interval`)**: Sources from `global_hcp` or `regime_hcp`, scaling unclipped half-width by $\lambda \in [0.01, 4.0]$ (plus $\{5, 6, 8, 12, 16\}$):

$$q_{\text{source}}(t) = U_{\text{source}}(t) - p(t) \tag{6.3}$$

$$L_\lambda(t) = \max(0,\ p(t) - \lambda q_{\text{source}}(t)), \quad U_\lambda(t) = p(t) + \lambda q_{\text{source}}(t) \tag{6.4}$$

Selected on FD002 validation data to minimize absolute difference from proposed validation MPIW (required ≤ 1.0 cycle): GBR selects `global_hcp` with $\lambda = 0.92$; LSTM selects `regime_hcp` with $\lambda = 0.90$.

2. **Widened Interval Family (`widened_interval`)**: Same scaling formula with $\lambda \geq 1.00$, minimizing validation cost: GBR selects `regime_hcp` with $\lambda = 1.00$; LSTM selects `global_hcp` with $\lambda = 1.00$.

3. **Point Safety Margin Family (`point_safety_margin`)**: Maintains iff $p \leq 1 + m$ for $m \in \{0, \ldots, 250\}$, minimizing validation cost: GBR selects $m = 5$; LSTM selects $m = 4$. As a point policy, width and conformal coverage metrics are designated `NOT_APPLICABLE`.

4. **Earlier Fixed Policy Family (`earlier_intervention`)**: Maintains iff $L_{\text{source}} \leq h$ for $h \in \{1, \ldots, 251\}$, minimizing validation cost: GBR selects `regime_hcp` with $h = 1$; LSTM selects `global_hcp` with $h = 1$.

All control selection executed on FD002 validation engines across seeds 2026–2030 under `primary_reference` at 90% coverage, completely isolated from calibration or held-out test data.

6.5 Statistical Inference Protocol and Decision Rules

Confirmatory hypothesis testing is executed strictly via paired engine-level cluster bootstrap resampling:

- **Cluster Unit**: The independent statistical unit is the physical engine unit ($i \in \{1, \ldots, N\}$). Individual cycles within an engine trajectory are never resampled independently. Seeds are averaged within engine clusters as descriptive training variability.

- **Bootstrap Replicates**: $B = 10{,}000$ paired bootstrap samples are drawn with replacement, stratified by engine lifetime quartiles (Q_1–Q_4), using `PCG64` PRNG with seed sequence `SeedSequence([2026, 6, dataset_number])`.

- **Equal Engine Weighting**: All fleet metrics (coverage, width, cost, failure rate) are computed with equal weighting per engine unit.

- **Missing-Group Bound Propagation**: If an eligible subgroup (≥ 10 engines) is absent in a bootstrap resample, its absolute coverage error is bounded by $[0.0, 0.90]$ at 90% nominal coverage and propagated conservatively.

- **Multiplicity Correction**: Family-wise error rate across RQ1–RQ4 is controlled via Bonferroni correction:

$$\alpha_{\text{conf}} = \frac{0.05}{4} = 0.0125. \tag{6.5}$$

  Confirmatory decision rules utilize 98.75% two-sided percentile confidence intervals (empirical percentiles 0.625% and 99.375%).

- **Intersection-Union Test (IUT)**: Benefit conditions pass if lower CI > margin; ceiling/noninferiority conditions pass if upper CI < ceiling. Equality with a boundary or degenerate inference is inconclusive. Failure safeguard uses observed fleet failure rates ($\Delta_{\text{fail}} \leq 0.0$). A hypothesis is decided as **Supported** iff all required criteria pass at the 98.75% level, **Unsupported** if any required condition fails, and otherwise **Inconclusive**.

Chapter 7

Results and Discussion

Status Notice: *The empirical evaluations and analytical tables reported in this chapter represent the pre-registered reporting templates for the confirmatory evaluation. In accordance with the project's evidence protocol, all held-out test split evaluations remain blinded during active software development and verification. Empirical entries are designated* `PENDING` *awaiting completion of experimental gates* UPD-02, UPD-03, *and* UPD-04. *No synthetic or unverified empirical numbers are reported.*

7.1 Point Degradation Predictor Evaluation

Point prediction accuracy on the held-out test engine splits is evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) computed with equal weighting across test engines. Table 7.1 outlines the evaluation structure.

Table 7.1: Point Prediction Performance on Held-Out Test Engines (Mean ± Std across 5 Seeds)

Dataset	**Architecture**	**RMSE (cycles)**	**MAE (cycles)**
FD001	GBR	*PENDING (UPD-02)*	*PENDING (UPD-02)*
FD001	LSTM	*PENDING (UPD-02)*	*PENDING (UPD-02)*
FD002	GBR	*PENDING (UPD-02)*	*PENDING (UPD-02)*
FD002	LSTM	*PENDING (UPD-02)*	*PENDING (UPD-02)*
FD004	GBR	*PENDING (UPD-02)*	*PENDING (UPD-02)*
FD004	LSTM	*PENDING (UPD-02)*	*PENDING (UPD-02)*

7.2 Conformal Interval Calibration Evaluation

Conformal prediction intervals are evaluated at nominal confidence level $1 - \alpha = 0.90$ across all 3 datasets $\times$ 2 models $\times$ 3 registered calibration methods (18 cells). Metrics comprise engine-weighted empirical coverage (Cov), mean interval width ($\bar{W}$), and Winkler interval score ($\text{IS}_{0.10}$). Table 7.2 summarizes the evaluation grid.

Table 7.2: Primary Interval Calibration Performance at Nominal 90% Confidence Level on Held-Out Test Splits ($3 \times 2 \times 3 = 18$ cells)

Dataset	Model	Method	Empirical Cov. (%)	Mean
FD001	GBR	Global HCP (`global_hcp`)	*PENDING*	*PEND*
FD001	GBR	Regime HCP (`regime_hcp`)*	*PENDING*	*PEND*
FD001	GBR	Decision-Aware HCP (`decision_aware_hcp`)	*PENDING*	*PEND*
FD001	LSTM	Global HCP (`global_hcp`)	*PENDING*	*PEND*
FD001	LSTM	Regime HCP (`regime_hcp`)*	*PENDING*	*PEND*
FD001	LSTM	Decision-Aware HCP (`decision_aware_hcp`)	*PENDING*	*PEND*
FD002	GBR	Global HCP (`global_hcp`)	*PENDING*	*PEND*
FD002	GBR	Regime HCP (`regime_hcp`)	*PENDING*	*PEND*
FD002	GBR	Decision-Aware HCP (`decision_aware_hcp`)	*PENDING*	*PEND*
FD002	LSTM	Global HCP (`global_hcp`)	*PENDING*	*PEND*
FD002	LSTM	Regime HCP (`regime_hcp`)	*PENDING*	*PEND*
FD002	LSTM	Decision-Aware HCP (`decision_aware_hcp`)	*PENDING*	*PEND*
FD004	GBR	Global HCP (`global_hcp`)	*PENDING*	*PEND*
FD004	GBR	Regime HCP (`regime_hcp`)	*PENDING*	*PEND*
FD004	GBR	Decision-Aware HCP (`decision_aware_hcp`)	*PENDING*	*PEND*
FD004	LSTM	Global HCP (`global_hcp`)	*PENDING*	*PEND*
FD004	LSTM	Regime HCP (`regime_hcp`)	*PENDING*	*PEND*
FD004	LSTM	Decision-Aware HCP (`decision_aware_hcp`)	*PENDING*	*PEND*

*On FD001 (single operating condition), `regime_hcp` operates with a single regime partition identical to global calibration.

7.3 Sequential Maintenance Policy Outcomes

Downstream maintenance outcomes under the primary economic scenario (`primary_reference`: $C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$ SCU; $L(t) \leq 1.0$ cycle trigger) are reported in Table 7.3 across the 18 primary method cells plus 8 pre-registered control configurations on FD002 (total 26 evaluation rows). Metrics include mean maintenance cost per engine ($\bar{C}$), fleet failure rate (P_{fail}), and mean wasted operational life ($\bar{W}$).

Table 7.3: Sequential Maintenance Policy Outcomes on Held-Out Test Engines (`primary_reference`: $C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$ SCU; 26 rows)

Dataset	Model	Policy / Calibration Strategy	Cost (SCU)	Failures
Panel A: Primary Registered Calibration Methods (18 Cells)				
FD001	GBR	Global HCP Lower Bound (`global_hcp`)	*PENDING*	*PEND*
FD001	GBR	Regime HCP Lower Bound (`regime_hcp`)	*PENDING*	*PEND*
FD001	GBR	Decision-Aware Lower Bound (`decision_aware_hcp`)	*PENDING*	*PEND*
FD001	LSTM	Global HCP Lower Bound (`global_hcp`)	*PENDING*	*PEND*
FD001	LSTM	Regime HCP Lower Bound (`regime_hcp`)	*PENDING*	*PEND*
FD001	LSTM	Decision-Aware Lower Bound (`decision_aware_hcp`)	*PENDING*	*PEND*
FD002	GBR	Global HCP Lower Bound (`global_hcp`)	*PENDING*	*PEND*
FD002	GBR	Regime HCP Lower Bound (`regime_hcp`)	*PENDING*	*PEND*
FD002	GBR	Decision-Aware Lower Bound (`decision_aware_hcp`)	*PENDING*	*PEND*
FD002	LSTM	Global HCP Lower Bound (`global_hcp`)	*PENDING*	*PEND*
FD002	LSTM	Regime HCP Lower Bound (`regime_hcp`)	*PENDING*	*PEND*
FD002	LSTM	Decision-Aware Lower Bound (`decision_aware_hcp`)	*PENDING*	*PEND*
FD004	GBR	Global HCP Lower Bound (`global_hcp`)	*PENDING*	*PEND*
FD004	GBR	Regime HCP Lower Bound (`regime_hcp`)	*PENDING*	*PEND*
FD004	GBR	Decision-Aware Lower Bound (`decision_aware_hcp`)	*PENDING*	*PEND*
FD004	LSTM	Global HCP Lower Bound (`global_hcp`)	*PENDING*	*PEND*
FD004	LSTM	Regime HCP Lower Bound (`regime_hcp`)	*PENDING*	*PEND*
FD004	LSTM	Decision-Aware Lower Bound (`decision_aware_hcp`)	*PENDING*	*PEND*
Panel B: Pre-Registered Matched Controls on FD002 (8 Cells)				
FD002	GBR	Matched-Width Interval (`global_hcp`, $\lambda = 0.92$)	*PENDING*	*PEND*
FD002	GBR	Widened Interval (`regime_hcp`, $\lambda = 1.00$)	*PENDING*	*PEND*
FD002	GBR	Point Safety Margin ($p \leq 1 + 5$)	*PENDING*	*PEND*
FD002	GBR	Earlier Intervention (`regime_hcp`, $h = 1$)	*PENDING*	*PEND*
FD002	LSTM	Matched-Width Interval (`regime_hcp`, $\lambda = 0.90$)	*PENDING*	*PEND*
FD002	LSTM	Widened Interval (`global_hcp`, $\lambda = 1.00$)	*PENDING*	*PEND*
FD002	LSTM	Point Safety Margin ($p \leq 1 + 4$)	*PENDING*	*PEND*
FD002	LSTM	Earlier Intervention (`global_hcp`, $h = 1$)	*PENDING*	*PEND*

For `point_safety_margin`, prediction interval width and conformal coverage metrics are designated `NOT_APPLICABLE`.

7.4 Hypothesis Decisions (RQ1–RQ4)

Confirmatory statistical inference is conducted via 10,000 paired engine-level cluster bootstrap resamples. Table 7.4 presents the formal decision matrix with Bonferroni-corrected 98.75% confidence intervals under the pre-registered Intersection-Union Test logic (ADR-018).

Table 7.4: Pre-Registered Confirmatory Hypothesis Decision Rules and Outcome Contrasts (ADR-018)

Research Question	**Contrast Metric & Scope**	**Pre-Registered Thresholds (98.75% CI)**	**Point Est.**
RQ1 (Utility & Coverage)	Cost improvement vs. `global_hcp` and `regime_hcp` on FD002 (GBR and LSTM)	$\Delta_{\text{cost}} > 0.20$ SCU; coverage errors < 0.05; $\Delta_{\text{fail}} \leq 0.0$	*PENDING*
RQ2 (Subgroup Reliability)	Worst-group coverage error contrast for regimes (C_1–C_6) and stages on FD002	$\Delta_{\text{error}} > 0.01$; worst error < 0.05; $D_{\text{width}} < 1.0$; $D_{\text{score}} < 1.0$ cycle	*PENDING*
RQ3 (Control Attribution)	Cost improvement vs. each of the 4 matched controls on FD002 (GBR and LSTM)	$\Delta_{\text{cost}} > 0.20$ SCU over every control; coverage < 0.05; $\Delta_{\text{fail}} \leq 0.0$	*PENDING*
RQ4 (Generalization)	Balanced 6-cell evaluation matrix ($w_k = 1/6$ each) across FD001, FD002, FD004	Aggregate: $\Delta_{\text{cost}} > 0.20, \Delta_{\text{error}} > 0.01$; Cell non-harm: $\Delta_{\text{cost}} > -0.20, \Delta_{\text{error}} > -0.01$	*PENDING*

Decision rules: An RQ is decided as `Supported` iff all required component conditions pass at the 98.75% level, `Unsupported` if any condition fails, and otherwise `Inconclusive`.

7.5 Methodological Discussion

7.5.1 Conservatism, Asymmetric Penalties, and Negative β

The scaling functions are defined by:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \tag{7.1}$$

$$a(p) = \sigma(p)\exp(\beta z(p)), \quad b(p) = \sigma(p)\exp(-\beta z(p)) \tag{7.2}$$

At fixed p, q, and γ, negative β decreases $a(p)$ relative to $\beta = 0$ and increases $b(p)$. The lower bound $L = \max(0, p - q_g a(p))$ is nondecreasing and the upper bound $U = p + q_g b(p)$ is nondecreasing. We do not claim $a < 1$ or $b > 1$ for every p, as the $\sigma(p)$ factor also influences their magnitude. Furthermore, this mathematical fact does not establish improved maintenance utility or earlier intervention; nonconformity thresholds q^* are fitted separately and utility must be evaluated empirically in sequential simulation.

7.5.2 Interpreting Potential Hypothesis Outcomes

The pre-registered protocol explicitly pre-commits to objective interpretation:

- **Supported**: All required component conditions pass at the 98.75% decision interval level, demonstrating statistically significant and practically meaningful superiority.
- **Inconclusive**: No condition fails, but at least one decision interval spans the practical threshold boundary, or inference is degenerate. In this case, no definitive superiority claim is made, and the result is cataloged as requiring expanded engine sample sizes.
- **Unsupported**: Any required component condition fails (e.g., upper CI below margin, lower CI above ceiling, or fleet failure rate increases). This outcome indicates that the pre-registered criteria were not supported under the specified experiment.

Final empirical conclusions remain pending held-out evaluation.

Chapter 8

Limitations, Threats to Validity, and Conclusion

8.1 Threats to Validity and Research Boundaries

To maintain rigorous scientific standards, the methodological conclusions and empirical findings of this research must be interpreted within explicitly defined boundaries. We identify eight primary threats to validity:

8.1.1 Benchmark Simulation Fidelity

The experimental investigation is conducted exclusively on the synthetic NASA C-MAPSS turbofan engine degradation simulation suite [1]. While C-MAPSS is the preeminent benchmark in PHM research, its degradation curves exhibit strictly monotonic wear profiles. Operational aircraft turbofan engines encounter complex operational realities—such as component washing, partial maintenance servicing, sensor drift, and intermittent fault reversibility—that are not modeled within C-MAPSS.

8.1.2 Synthetic Cost and Economic Assumptions

The discrete-event maintenance simulation operates on study-defined cost preferences under the primary reference scenario ($C_{\text{PM}} = 1.0$ SCU, $C_{\text{FAIL}} = 5.0$ SCU, $C_{\text{WASTE}} = 0.20$ SCU/cycle). Although these parameters realistically reflect the order-of-magnitude cost disparity between scheduled overhauls and in-service engine failures, they represent benchmark simulation preferences rather than empirical airline accounting constants.

8.1.3 Finite Fleet Engine Counts

Across the complete C-MAPSS benchmark suite, 609 engines are evaluated, with 122 engines allocated to the held-out test splits. While this sample size is substantial for

academic prognostics benchmarks, statistical power in paired cluster-bootstrap inference across rare flight regimes remains limited.

8.1.4 Temporal Autocorrelation and Conformal Exchangeability

Hierarchical conformal theory provides marginal predictive coverage under its stated sampling and exchangeability assumptions. IntelliTwin uses an engine-weighted construction motivated by that theory. Applicability of those assumptions to ordered, lifetime-stratified degradation trajectories is not established merely by weighting engines equally; reliability is therefore evaluated empirically under the frozen protocol. Furthermore, downstream sequential stopping times are not theoretically guaranteed by split conformal guarantees.

8.1.5 Theoretical Conformal Coverage Guarantees

Conformal calibration guarantees marginal coverage over the joint distribution of calibration and test engines. It does not provide theoretical guarantees of exact pointwise conditional coverage or sequential stopping-time coverage along an individual engine's degradation path.

8.1.6 Capped Training Targets vs. Uncapped Evaluation

Base models are trained using piecewise-capped RUL targets ($R_{\max} = 125$ cycles) as a modeling convention. In contrast, calibration, validation, and held-out test splits evaluate uncapped physical RUL. This distributional discrepancy must be accounted for during early-lifecycle inference.

8.1.7 Rare-Group Support and Fallback Protocols

In complex multi-regime environments (FD002 and FD004), rare flight conditions may contain fewer than the required support threshold $m_g = \max(10, \lceil 1/\alpha \rceil - 1)$ distinct contributing calibration engines (10 engines for 80% and 90% coverage; 19 engines for 95% coverage). When a regime lacks sufficient support, the framework executes a deterministic fallback to the pooled global quantile q^*_{global} for the same method and coverage level, recording fallback.

8.1.8 Research Prototype Boundaries

The software is an academic research prototype evaluating an offline, trace-driven simulation replay framework; it makes no claim of production readiness, aircraft certification,

physical hardware integration, or independently established end-to-end operational reproducibility.

8.2 Summary of Thesis Contributions

This dissertation has established a comprehensive, scientifically audited research framework for industrial turbofan prognostic maintenance:

1. **Methodological Synthesis**: Integration of modern base degradation predictors (GBR, LSTM) with hierarchical, condition-adaptive, and asymmetric conformal calibration.

2. **Rigorous Governance**: Complete engine-level disjoint data partitioning, eliminating temporal data leakage across 609 engine units.

3. **Principled Baseline Controls**: Formal evaluation against four named control strategies across three distinct families, isolating the true operational value of calibrated uncertainty.

4. **Pre-Registered Statistical Protocols**: Family-wise error rate control via Bonferroni-corrected 98.75% confidence intervals generated through 10,000 paired engine-cluster bootstrap resamples.

8.3 Path Forward to Final Experimental Completion

The theoretical formulation, software implementation, calibration registries, and evaluation protocols are fully specified, frozen, and audited. The remaining milestones comprise the execution of verification repair gates (`UPD-01`), held-out test split inference (`UPD-02`), discrete-event fleet simulation (`UPD-03`), confirmatory hypothesis resolution (`UPD-04`), demonstrator integration (`UPD-06`), and independent reproducibility verification (`UPD-07`). Upon completion of these gates, empirical findings will be incorporated directly into this dissertation.

Appendix A

Reproducibility Protocol and Artifact Registry

A.1 Replication Charter and Codebase Structure

The IntelliTwin framework is engineered for scientific auditability and reproducibility. All code, configuration files, dataset split manifests, and audit reports are version-controlled in the official repository:

`https://github.com/sik04/AI-Driven-Digital-Twin-Predictive-Maintainence`

The canonical reviewed reference commit for all methodology, architecture decisions, and frozen parameters described in this dissertation is:

`bb9e34579343881a831e3f052608ee15e183df76`

A.2 Software Environment and Dependencies

The framework defines declared package constraints via the `[project.optional-dependencies] research` extra in `pyproject.toml`:

- `numpy` $\geq 1.24.0$;
- `pandas` $\geq 2.0.0$;
- `scikit-learn` $\geq 1.2.0$;
- `torch` ≥ 2.2;
- `PyYAML` $\geq 6, < 7$.

Note that `lightgbm` is not a project dependency; base GBR predictors are implemented strictly via `sklearn.ensemble.HistGradientBoostingRegressor`.

We distinguish declared dependency constraints from recorded runtime environments and environments tested by automated Continuous Integration (CI). CI currently validates Python 3.10; universal compatibility across every unconstrained Python version ≥ 3.10 is not claimed.

A.3 Committed Pre-Test Audit Findings

Pretest verification artifacts committed under PR #25 establish the current verification status:

- **Full Audit Report** (`experiments/pretest_verification_v2/step08_full_audit.json`): Documents all 6 mandatory audit checks executed and passed. Evaluates 60 base-model partition predictions across the `validation` and `calibration` partitions (not training and validation), achieving an exact maximum absolute difference of 0.0.
- **Pretest Audit Manifest** (`experiments/pretest_verification_v2/pretest_audit.json`): Formally confirms status `PASS` and `PRETEST_READY`.
- **Baseline Inventory** (`experiments/pretest_verification_v2/baseline_manifest.json`): Delineates 652 tracked, protected Git artifacts from 72 historical local-only search checkpoints.
- **Held-Out Evaluation**: Official held-out test split evaluation remains `NOT_STARTED`, and all confirmatory research question outcomes remain `PENDING`.

A.4 Execution Commands and Audit Pipelines

The repository provides verified scripts for auditing models and artifacts (verified via source inspection):

1. **Data Pipeline Test Suite**:

   ```
   pytest tests/test_data_pipeline.py
   ```

2. **Base Model Registry Audit**:

   ```
   python scripts/audit_step07_base_models.py --reproduce-predictions
   ```

3. **Conformal Calibration Audit**:

```
python scripts/audit_step08_calibration.py
```

4. **Pre-Test Confirmatory Freeze Verification**:

```
python scripts/audit_step09_freeze.py
```

5. **Smoke Test Validation Execution**:

```
python scripts/run_step09_validation_smoke.py
```

A.5 Demarcation of Clean-Environment Verification

Reproducibility Notice: *The software is an academic research prototype. Independent clean-environment reproducibility (executing complete model retraining and calibration fitting from raw data in an isolated container) has not yet been executed by this agent and is tracked as an independent milestone under* `UPD-07`.

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